

is interesting because it may be that GPT-4’s forecasts are especially good, or especially bad, at predicting questions that resolve very quickly compared to those that do not. See Figure 4 for a scatterplot illustrating accuracy and question duration with linear fits. Our results show that we do not detect statistically significant effects of duration, $ps > .36$.

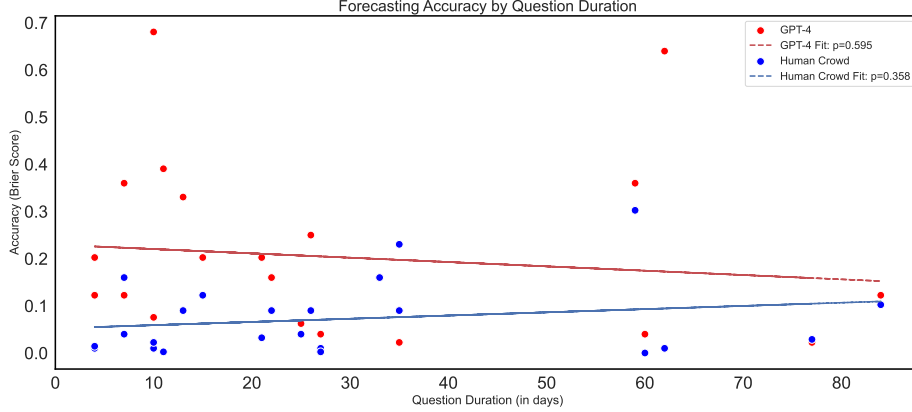


Figure 4: Scatterplot and linear fit for the relationship between forecasting accuracy (Brier score) and duration (in days) by condition.

To test this question directly, we conducted an Analysis of Covariance (ANCOVA) to assess whether the relationship between accuracy and question duration differs between GPT-4’s forecasts and those made by human forecasters. The main effect of the method (human crowd vs. GPT-4) was statistically significant, $F(1, 42) = 9.38, p = .004$, indicating a notable difference in accuracy between the two forecasting methods. However, the main effect of duration was not significant, $F(1, 42) = 0.017, p = .898$, suggesting that question duration did not significantly affect the accuracy of forecasts. Moreover, the interaction between method and duration was also not significant, $F(1, 42) = 0.75, p = .392$, suggesting that the relationship between question duration and accuracy did not significantly vary between GPT-4 and the human crowd.

Additionally, there are also a variety of aggregation techniques that may prove useful in combining the aggregate human forecast and GPT-4’s forecast. We report the exploratory results of a Bayesian Model Averaging (BMA) approach, which takes into account the uncertainty across models prior to combining the predictions from multiple models. We first calculate the likelihoods of each model (.82 for GPT-4 and .93 for the human crowd) before normalizing them and computing the posterior model probabilities. With these, we have the following weighted average:

$$\text{Brier}_{\text{BMA}} = (0.467 \cdot \text{Brier}_{\text{GPT-4}}) + (0.533 \cdot \text{Brier}_{\text{Human Crowd}}). \quad (2)$$

We compute $\text{Brier}_{\text{BMA}} = .13$ ($SD = .09$), which is significantly more accurate than the random baseline, $t(22) = -5.91, p < .001$.

We have presented this simple aggregation as an instance of how LLM predictions may feed into future aggregation approaches. We point out that given our finding of poor prediction accuracy, current models like GPT-4 may be unlikely to make worthwhile additions to aggregation algorithms. However, this may change with advances in LLM capabilities and LLM-forecasting technology.

4 Discussion

Our findings from entering GPT-4 into a real-world forecasting tournament on the Metaculus platform suggest that even this state-of-the-art LLM has unimpressive forecasting capabilities. Despite being prompted with established superforecasting techniques and best-practice prompting approaches, GPT-4 was heavily outperformed by the forecasts of the human crowd, and did not even outperform a no-information baseline of predicting 50% on every question. The robustness of this finding is suggested by the fact that the question set was diverse, drawing on a wide variety of topics like Big Tech, U.S. politics, viral outbreaks, and the Ukraine conflict. We argue that our data also provide

additional evidence for the predictive power of human crowd forecasting competitions, as their accuracy was impressive throughout the questions studied here. Our results thus suggest that current LLMs may not yet perform well in a variety of real-world prediction tasks that would necessitate probabilistic foresight.

One potential contributing factor to GPT-4’s inadequate forecasting capability is the fact that its training data is subject to a knowledge cutoff after a certain point in time.²⁸ In contrast to human forecasters who keep up-to-date with recent events, GPT-4 is not updated with events that occur post-training. Given the dynamic nature of many world events, GPT-4’s lack of real-time knowledge updating can be a significant barrier to accurate forecasting. Our design tried to address this concern by inputting the question’s background information presented on Metaculus to the LLM, so that it would not be wholly unaware of a potentially novel context. While this does not fully bring the model up to speed with human forecasters that seek out prediction competition on Metaculus, it does help mitigate the information gap posed by the knowledge cutoff. Additional ways to mitigate the limitations to forecasting posed by LLMs’ knowledge cutoff would be fruitful in future studies.¹⁸

OpenAI’s mission is to create “highly autonomous systems that outperform humans at most economically valuable work.”²⁹ Whether this AI-led future is on track to occur will likely be in large part determined by how capable LLMs—and AI systems in general—turn out to be at economically relevant tasks,³¹ especially without human hand-holding. Forecasting is a task of especially high economic relevance, especially for a large proportion of white-collar fields—such as business, policy, and law—that rely heavily on accurate predictions across a myriad of contexts. As of now, our results suggest that even state-of-the-art AI systems are not yet posed to replace human expertise in these areas, due to the inadequate forecasting capabilities of these systems. However, it will be especially important to closely monitor AI capability advances in the domain of accurate forecasting, as foresight will remain an essential skill for autonomous systems.

Another relevant implication of AI forecasting capabilities—or lack thereof—is the risk arising from AI systems that are proficient at long-term planning. An AI system with robust long-term planning capabilities would be able to presciently pursue their goal, which can be a sizable and potentially catastrophic danger if the goal happens to be incompatible with the well-being of humans²⁶ (e.g., the goal of engineering a pandemic that kills as many people as possible over the long run). Proficiency in long-term planning requires the ability to accurately forecast future scenarios. For many real-world tasks, such accurate forecasting is a highly complex endeavor, at which even many (but not all) humans arguably perform poorly.³⁴ Our finding that GPT-4 has particularly poor forecasting capabilities bolsters the case that the threat of an AI system planning in the long term against human interests remains presently low. However, this should not be taken as a reason to be complacent. The pace at which AI capabilities advance suggests that it remains crucial to continually monitor the progression of AI systems in terms of their forecasting abilities to ensure that their development remains robustly safe.

Our results raise several avenues for further research. First, it may be argued that our findings might, at least in part, be explained by the human comparison group’s ability to harness the wisdom of the crowd,¹⁴ whereas the LLM prediction might be best understood as a single forecast only. One counterargument to this claim is that LLM predictions themselves may draw on a wisdom-of-the-crowd effect from their large and multifaceted training dataset. Future research on how to employ a larger ensemble of LLM forecasters that draw on diverse inputs, training datasets, prompts, temperature values, and other variations would be fruitful.

Second, there may be merit in implementing LLM forecasters that can access the internet for information and autonomously update their forecasts. While the design and implementation of such a setup pose challenges, particularly as most such design choices require human input which may introduce bias, it remains a noteworthy endeavor. Internet access may also help mitigate the design weakness outlined earlier, where the temporal distance between the training data cutoff and the forecasting question’s start may impair contextual understanding of the question details.

Third, we outlined a simple Bayesian model-averaging approach for aggregating human and machine forecasts. Future work may delve deeper into alternative ways to combine such forecasts, such as exploring ways of using LLM forecasts in extremizing algorithms or other ways of selectively choosing to favor or disfavor LLM forecasts depending on their eventual performance and comparative strengths and weaknesses. However, it is worth pointing out that given the current poor prediction

performance, such approaches probably only become worthwhile once LLM foresight abilities are significantly improved.

Finally, it would be productive to investigate the integration of human and LLM forecasts, such as by examining whether hybrid forecasting models outperform the standalone baselines. Such hybrid forecasters (i.e., human forecasters that rely upon LLM outputs to make predictions) may be able to combine the respective strengths of both human cognition and LLM cognition, which open up the potential for new forecasting techniques based on the augmentation—rather than replacement—of humans by AI.³

Our data suggest a clear weakness in an otherwise impressive catalogue of LLM capabilities: predicting the future. This highlights a number of technical challenges and future research directions for how LLMs may be harnessed for forecasting, as well as for various real-world tasks that require or are helped by robust forecasting capabilities. Our results suggest that the wisdom of human crowds remains a powerful tool in providing probabilistic forecasts about the future. This context of a forecasting tournament may also prove to be an especially fruitful environment for probing the degree to which LLMs are capable of generalized reasoning and prediction.

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Data availability

Our pre-registration and analysis plan can be found in our Open Science Foundation database: https://osf.io/tfbvd/?view_only=bfc6d61152654e3b84c36446f1858fb2

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